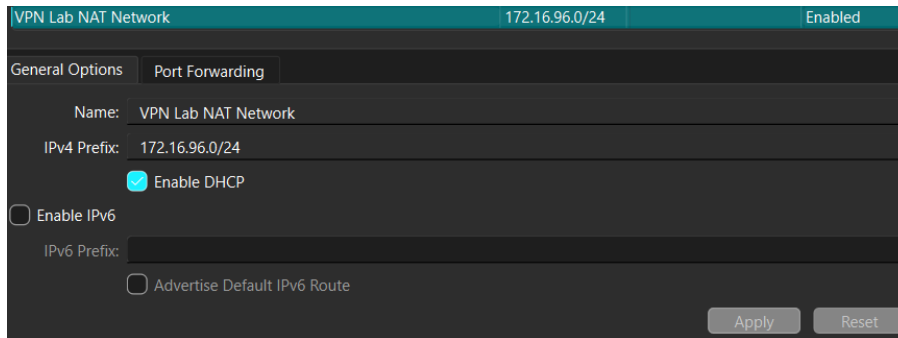


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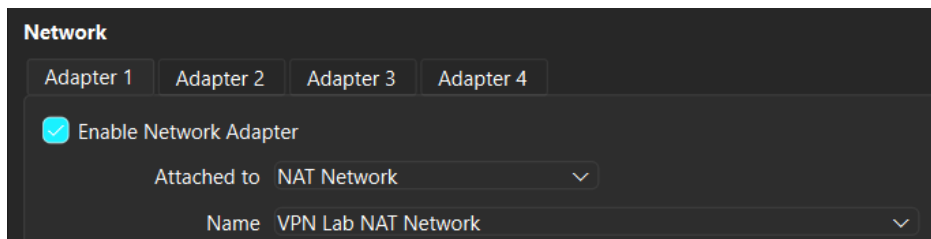
Configure setup:

Create the network in VirtualBox: 172.16.96.0/24



The screenshot shows the 'VPN Lab NAT Network' configuration window. The 'General Options' tab is selected. The 'Name' is 'VPN Lab NAT Network', the 'IPv4 Prefix' is '172.16.96.0/24', and 'Enable DHCP' is checked. 'Enable IPv6' is unchecked, and 'Advertise Default IPv6 Route' is also unchecked. 'Apply' and 'Reset' buttons are at the bottom right.

Add the machines to the network (Ub1, Ub2 and Kali):



The screenshot shows the 'Network' configuration window. The 'Adapter 1' tab is selected. 'Enable Network Adapter' is checked. The 'Attached to' dropdown is set to 'NAT Network', and the 'Name' dropdown is set to 'VPN Lab NAT Network'.

The ips are:

172.16.96.3/24 -> Kali (Server) (stud-stud)

172.16.96.4/24 -> Ubuntu1 (Client) (student-student)

172.16.96.5/24 -> Ubuntu2 (Client) (student-student)

Server: Necessary files

File	Purpose
server.conf	Main OpenVPN server configuration file
server.crt	Server certificate (signed by CA)
server.key	Server private key
ca.crt	Certificate Authority certificate (used to verify clients)
ta.key	TLS authentication HMAC key (same key shared with clients)

```
(stud@kali-vm)-[/etc/openvpn/server]
$ ls
ca.crt  server.conf  server.crt  server.key  ta.key
```

server.conf:

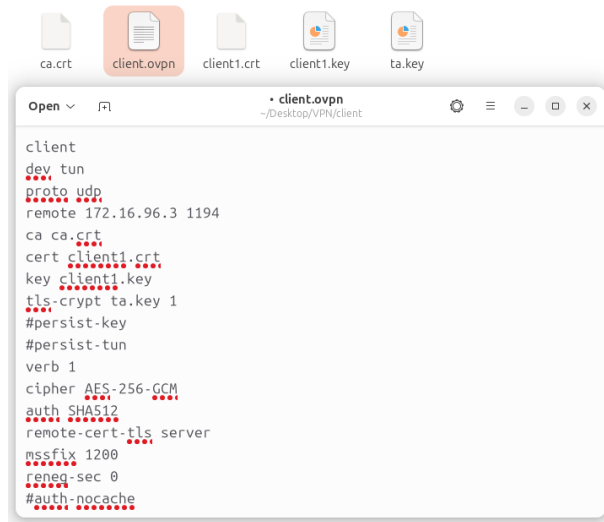
```
(stud@kali-vm)-[/etc/openvpn/server]
$ sudo systemctl status openvpn-server@server.service
● openvpn-server@server.service - OpenVPN service for server
   Loaded: loaded (/usr/lib/systemd/system/openvpn-server@.service; enabled; preset: disabled)
   Active: active (running) since Sun 2025-12-07 22:27:01 CET; 1min 15s ago
     Docs: man:openvpn(8)
           https://openvpn.net/community-resources/reference-manual-for-openvpn-2-6/
           https://community.openvpn.net/openvpn/wiki/HOWTO
   Main PID: 52142 (openvpn)
   Status: "Initialization Sequence Completed"
     Tasks: 1 (limit: 4610)
    Memory: 1.9M (peak: 2.1M)
       CPU: 16ms
   CGroup: /system.slice/system-openvpn@x2dsrver.slice/openvpn-server@server.service
           └─52142 /usr/sbin/openvpn --status /run/openvpn-server/status-server.log --status-version 2 --suppress
Dec 07 22:27:01 kali-vm openvpn[52142]: Could not determine IPv4/IPv6 protocol. Using AF_INET
Dec 07 22:27:01 kali-vm openvpn[52142]: Socket Buffers: R=[212992→212992] S=[212992→212992]
Dec 07 22:27:01 kali-vm openvpn[52142]: UDPv4 link local (bound): [AF_INET][undef]:1194
Dec 07 22:27:01 kali-vm openvpn[52142]: UDPv4 link remote: [AF_UNSPEC]
Dec 07 22:27:01 kali-vm openvpn[52142]: UID set to nobody
Dec 07 22:27:01 kali-vm openvpn[52142]: GID set to nogroup
Dec 07 22:27:01 kali-vm openvpn[52142]: Capabilities retained: CAP_NET_ADMIN
Dec 07 22:27:01 kali-vm openvpn[52142]: MULTI: multi_init called, r=256 v=256
Dec 07 22:27:01 kali-vm openvpn[52142]: IFCONFIG POOL IPv4: base=10.88.88.4 size=62
Dec 07 22:27:01 kali-vm openvpn[52142]: Initialization Sequence Completed
```

File	Purpose
client.ovpn	Main OpenVPN client configuration file
clientX.crt	Client certificate (client1.crt or client2.crt)
clientX.key	Client private key (client1.key or client2.key)
ca.crt	CA certificate (same as server, to verify server identity)
ta.key	TLS authentication HMAC key (same as server)

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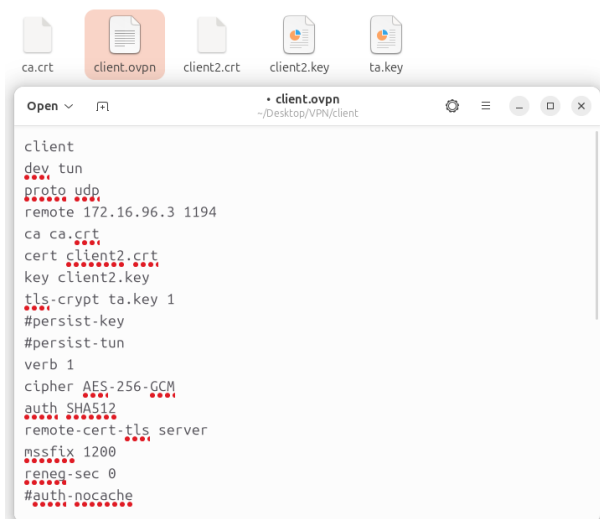
Client1:



Start connection: > sudo openvpn --config /home/student/Desktop/VPN/client/client.ovpn

```
student@student-ubuntu:~$ cd Desktop/VPN/client
student@student-ubuntu:~/Desktop/VPN/client$ sudo openvpn --config /home/student/Desktop/VPN/client/client.ovpn
[sudo] password for student:
2025-12-07 22:43:44 WARNING: file 'client1.key' is group or others accessible
2025-12-07 22:43:44 WARNING: file 'ta.key' is group or others accessible
2025-12-07 22:43:44 OpenVPN 2.6.9 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
2025-12-07 22:43:44 library versions: OpenSSL 3.0.13 30 Jan 2024, LZO 2.10
2025-12-07 22:43:44 DCO version: N/A
2025-12-07 22:43:44 TCP/UDP: Preserving recently used remote address: [AF_INET]172.16.96.3:1194
2025-12-07 22:43:44 UDPv4 link local: (not bound)
2025-12-07 22:43:44 UDPv4 link remote: [AF_INET]172.16.96.3:1194
2025-12-07 22:43:44 [alvaroscar] Peer Connection Initiated with [AF_INET]172.16.96.3:1194
2025-12-07 22:43:44 TUN/TAP device tun0 opened
2025-12-07 22:43:44 net_iface_mtu_set: mtu 1500 for tun0
2025-12-07 22:43:44 net_iface_up: set tun0 up
2025-12-07 22:43:44 net_addr_ptp_v4_add: 10.88.88.6 peer 10.88.88.5 dev tun0
2025-12-07 22:43:44 Initialization Sequence Completed
```

Client2:



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And start connection:

```
student@student-ubuntu:~$ cd Desktop/VPN/client
student@student-ubuntu:~/Desktop/VPN/client$ ls
ca.crt  client2.crt  client2.key  client.ovpn  ta.key
student@student-ubuntu:~/Desktop/VPN/client$ sudo openvpn --config /home/student/Desktop/VPN/client/client.ovpn
[sudo] password for student:
2025-12-07 22:51:59 WARNING: file 'client2.key' is group or others accessible
2025-12-07 22:51:59 WARNING: file 'ta.key' is group or others accessible
2025-12-07 22:51:59 OpenVPN 2.6.9 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [
DCO]
2025-12-07 22:51:59 library versions: OpenSSL 3.0.13 30 Jan 2024, LZO 2.10
2025-12-07 22:51:59 DCO version: N/A
2025-12-07 22:51:59 TCP/UDP: Preserving recently used remote address: [AF_INET]172.16.96.3:1194
2025-12-07 22:51:59 UDPv4 link local: (not bound)
2025-12-07 22:51:59 UDPv4 link remote: [AF_INET]172.16.96.3:1194
2025-12-07 22:51:59 [alvaroscar] Peer Connection Initiated with [AF_INET]172.16.96.3:1194
2025-12-07 22:51:59 TUN/TAP device tun0 opened
2025-12-07 22:51:59 net_iface_mtu_set: mtu 1500 for tun0
2025-12-07 22:51:59 net_iface_up: set tun0 up
2025-12-07 22:51:59 net_addr_pton_v4_add: 10.88.88.10 peer 10.88.88.9 dev tun0
2025-12-07 22:51:59 Initialization Sequence Completed
```

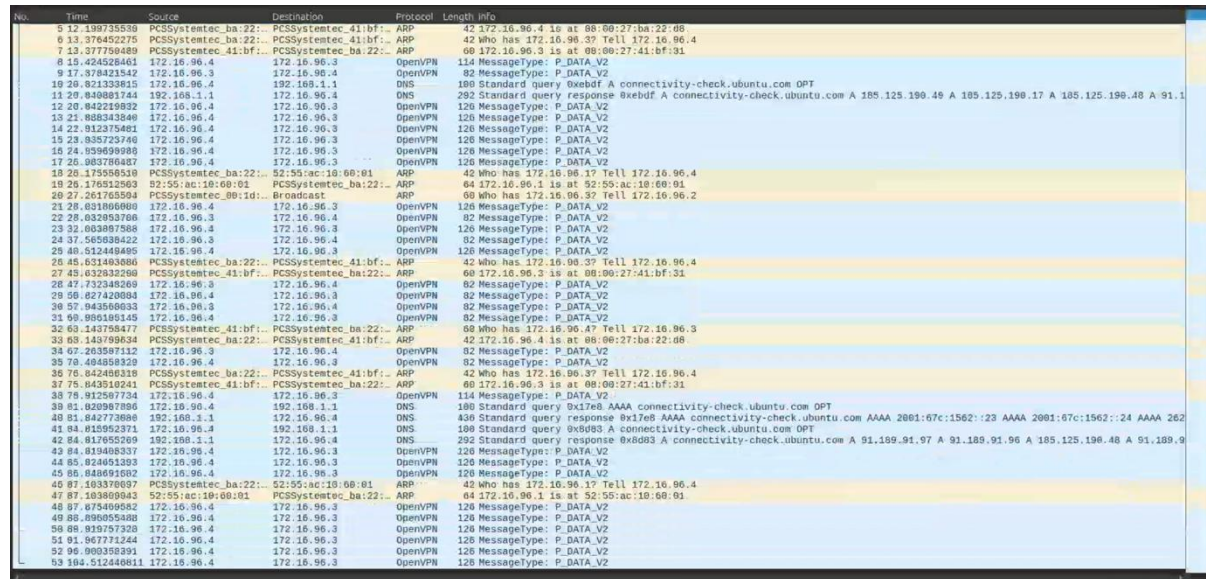

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1 TLS communications:

The point aims to analyze the TLS 1.2/TLS 1.3 communications. Get acquainted with TLS/SSL versions 1.2 and 1.3, find the advantages and disadvantages of each version.

1.1 Capture the network traffic on the physical network interface (usually, the name starts with en...):

Physical Network: eth0



No.	Time	Source	Destination	Protocol	Length	Info
5	12.199735539	PCSSystemtec_ba:22::	PCSSystemtec_41:bf::	ARP	42	172.16.96.4 is at 08:00:27:ba:22:08
6	13.376422775	PCSSystemtec_ba:22::	PCSSystemtec_41:bf::	ARP	42	Who has 172.16.96.3 Tell 172.16.96.4
7	13.377750489	PCSSystemtec_41:bf::	PCSSystemtec_ba:22::	ARP	60	172.16.96.3 is at 08:00:27:41:bf:31
8	15.424528461	172.16.96.4	172.16.96.3	OpenVPN	114	MessageType: P_DATA_V2
9	17.978421542	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V2
10	20.821333815	172.16.96.4	192.168.1.1	DNS	180	Standard query 0xebdf A connectivity-check.ubuntu.com OPT
11	20.840881744	192.168.1.1	172.16.96.4	DNS	292	Standard query response 0xebdf A connectivity-check.ubuntu.com A 185.125.190.46 A 185.125.190.17 A 185.125.190.48 A 91.1
12	20.842219832	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
13	21.808243546	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
14	22.912375461	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
15	23.835725740	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
16	24.958599080	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
17	26.983766487	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
18	26.175958510	PCSSystemtec_ba:22::	52:55:ac:10:60:01	ARP	42	Who has 172.16.96.17 Tell 172.16.96.4
19	26.176912563	52:55:ac:10:60:01	PCSSystemtec_ba:22::	ARP	64	172.16.96.1 is at 52:55:ac:10:60:01
20	27.261765594	PCSSystemtec_00:1d::	Broadcast	ARP	60	Who has 172.16.96.37 Tell 172.16.96.2
21	28.031066080	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
22	28.032937686	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V2
23	32.003897588	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
24	37.565938422	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V2
25	40.512448495	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
26	45.031483886	PCSSystemtec_ba:22::	PCSSystemtec_41:bf::	ARP	42	Who has 172.16.96.37 Tell 172.16.96.4
27	45.032832260	PCSSystemtec_41:bf::	PCSSystemtec_ba:22::	ARP	60	172.16.96.3 is at 08:00:27:41:bf:31
28	47.732345260	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V2
29	50.627420084	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V2
30	57.943560633	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V2
31	59.905185145	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V2
32	60.145758477	PCSSystemtec_41:bf::	PCSSystemtec_ba:22::	ARP	60	Who has 172.16.96.47 Tell 172.16.96.3
33	63.143798634	PCSSystemtec_ba:22::	PCSSystemtec_41:bf::	ARP	42	172.16.96.4 is at 08:00:27:ba:22:08
34	67.263587112	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V2
35	70.404585329	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V2
36	76.842465319	PCSSystemtec_ba:22::	PCSSystemtec_41:bf::	ARP	42	Who has 172.16.96.37 Tell 172.16.96.4
37	76.843910241	PCSSystemtec_41:bf::	PCSSystemtec_ba:22::	ARP	60	172.16.96.3 is at 08:00:27:41:bf:31
38	79.912507734	172.16.96.4	172.16.96.3	OpenVPN	114	MessageType: P_DATA_V2
39	81.020897886	172.16.96.4	192.168.1.1	DNS	180	Standard query 0x17e8 AAAA connectivity-check.ubuntu.com OPT
40	81.842773886	192.168.1.1	172.16.96.4	DNS	436	Standard query response 0x17e8 AAAA connectivity-check.ubuntu.com AAAA 2001:67c:1562::23 AAAA 2001:67c:1562::24 AAAA 262
41	84.815952371	172.16.96.4	192.168.1.1	DNS	180	Standard query 0x5083 A connectivity-check.ubuntu.com OPT
42	84.817955269	192.168.1.1	172.16.96.4	DNS	292	Standard query response 0x5083 A connectivity-check.ubuntu.com A 91.189.91.97 A 91.189.91.96 A 185.125.190.46 A 91.189.9
43	84.819408337	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
44	85.824651393	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
45	86.848491602	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
46	87.103376697	PCSSystemtec_ba:22::	52:55:ac:10:60:01	ARP	42	Who has 172.16.96.17 Tell 172.16.96.4
47	87.103898843	52:55:ac:10:60:01	PCSSystemtec_ba:22::	ARP	64	172.16.96.1 is at 52:55:ac:10:60:01
48	87.875496562	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
49	89.890655480	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
50	89.89257320	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
51	91.967771244	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
52	96.900335391	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
53	104.512448611	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2

In the network capture taken on the physical network interface, most of the packets are identified as **OpenVPN** traffic (e.g., MessageType: P_DATA_V2). This behavior is expected when the client is actively connected to the VPN. Once the VPN tunnel is established, all outgoing and incoming traffic from the system is encapsulated inside the encrypted OpenVPN tunnel before it is transmitted over the physical interface.

For this reason, the physical interface does not display the original application-layer traffic (such as HTTPS/TLS handshakes, HTTP requests, or any plaintext data). Instead, it only shows the encrypted VPN packets that transport internal communication. The content of these packets cannot be inspected, as OpenVPN encrypts the payload using the configured cipher and authentication algorithms.

Additionally, some DNS queries (e.g., connectivity checks from Ubuntu) and ARP messages may appear in the capture. These packets are generated by the operating system for normal network operation and are not part of the VPN's encrypted payload. They are visible because they occur outside the VPN tunnel

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1.2 Filter only the records for TLS/SSL communication. Filter expression for TLS/SSL in Wireshark: ssl.record.version == 0x0303

When the TLS/SSL filter is applied in Wireshark (e.g., `ssl.record.version == 0x0303`), the capture initially shows no traffic. This occurs because, while the VPN is active, all outgoing data—including TLS sessions—is encrypted and encapsulated inside OpenVPN packets before reaching the physical network interface. Since the TLS traffic is hidden inside the VPN tunnel, Wireshark cannot detect or classify these packets as TLS, and the filter returns an empty result.

However, once a web browser is opened and an HTTPS request is made, and the correct filter (`tls`) is applied on the appropriate interface, the capture begins to display TLSv1.2/TLSv1.3 handshake messages, such as Server Hello, Certificate, Key Exchange, and later Application Data, which represents encrypted HTTP content. This filtered traffic corresponds to the actual TLS communication generated by the browser during the secure web session.

No.	Time	Source	Destination	Protocol	Length	Info
1120	15.692004143	172.16.96.4	3.254.239.146	TLSv1.3	70	Application Data
1121	15.692004287	172.16.96.4	3.254.239.146	TCP	54	60440 - 60440 [ACK] Seq=6278 Ack=6625 Win=0 Len=0
1122	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1123	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1124	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1125	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1126	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1127	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1128	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1129	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1130	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1131	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1132	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1133	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1134	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1135	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1136	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1137	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1138	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1139	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1140	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1141	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1142	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1143	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1144	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1145	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1146	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1147	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1148	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1149	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1150	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1151	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1152	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1153	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1154	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1155	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1156	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1157	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1158	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1159	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1160	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1161	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1162	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1163	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1164	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1165	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1166	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1167	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1168	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1169	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1170	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1171	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1172	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1173	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1174	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1175	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1176	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1177	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1178	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1179	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1180	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1181	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1182	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1183	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1184	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1185	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1186	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1187	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1188	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1189	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1190	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1191	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1192	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1193	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1194	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1195	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1196	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1197	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1198	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0
1199	15.692004284	3.254.239.146	172.16.96.4	TCP	60	443 - 60440 [ACK] Seq=6425 Ack=1980 Win=0 Len=0

And filtered:

No.	Time	Source	Destination	Protocol	Length	Info
17	4.506715035	151.101.133.91	172.16.96.4	TLSv1.2	1510	Server Hello
18	4.506755630	151.101.133.91	172.16.96.4	TLSv1.2	1494	Certificate
21	4.506961025	151.101.133.91	172.16.96.4	TLSv1.2	221	Server Key Exchange, Server Hello Done
23	4.513247741	151.101.133.91	172.16.96.4	TLSv1.2	1510	Server Hello
24	4.513247967	151.101.133.91	172.16.96.4	TLSv1.2	1494	Certificate
27	4.514909005	151.101.133.91	172.16.96.4	TLSv1.2	221	Server Key Exchange, Server Hello Done
29	4.516541356	172.16.96.4	151.101.133.91	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
31	4.520169646	172.16.96.4	151.101.133.91	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
33	4.533959819	151.101.133.91	172.16.96.4	TLSv1.2	312	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
34	4.538622241	151.101.133.91	172.16.96.4	TLSv1.2	312	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
35	4.545174804	172.16.96.4	151.101.133.91	TLSv1.2	153	Application Data
36	4.546227996	172.16.96.4	151.101.133.91	TLSv1.2	346	Application Data
38	4.555439705	172.16.96.4	151.101.133.91	TLSv1.2	170	Application Data
39	4.556931452	172.16.96.4	151.101.133.91	TLSv1.2	281	Application Data
42	4.562336551	151.101.133.91	172.16.96.4	TLSv1.2	126	Application Data
43	4.562631596	172.16.96.4	151.101.133.91	TLSv1.2	92	Application Data
44	4.563174219	151.101.133.91	172.16.96.4	TLSv1.2	747	Application Data, Application Data
46	4.572264952	151.101.133.91	172.16.96.4	TLSv1.2	126	Application Data
47	4.572634772	172.16.96.4	151.101.133.91	TLSv1.2	85	Encrypted Alert
48	4.573662549	151.101.133.91	172.16.96.4	TLSv1.2	276	Application Data
49	4.573662829	151.101.133.91	172.16.96.4	TLSv1.2	258	Application Data
53	4.581362473	151.101.133.91	172.16.96.4	TLSv1.2	89	Encrypted Alert
58	4.595274937	172.16.96.4	151.101.133.91	TLSv1.2	218	Application Data
60	4.579997339	151.101.133.91	172.16.96.4	TLSv1.2	346	Application Data
61	4.579997744	151.101.133.91	172.16.96.4	TLSv1.2	770	Application Data
63	4.581364616	172.16.96.4	151.101.133.91	TLSv1.2	180	Application Data
65	4.581364616	151.101.133.91	172.16.96.4	TLSv1.2	1191	Application Data
70	4.582221601	34.160.144.191	172.16.96.4	TLSv1.2	1468	Server Hello
79	4.582221601	34.160.144.191	172.16.96.4	TLSv1.2	1494	Certificate
82	4.583508059	34.160.144.191	172.16.96.4	TLSv1.2	222	Server Key Exchange, Server Hello Done
84	4.581676551	34.160.144.191	172.16.96.4	TLSv1.2	1468	Server Hello
85	4.581677662	34.160.144.191	172.16.96.4	TLSv1.2	1494	Certificate
89	4.582522262	34.160.144.191	172.16.96.4	TLSv1.2	222	Server Key Exchange, Server Hello Done
92	4.582335674	172.16.96.4	34.160.144.191	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
93	4.582334054	172.16.96.4	34.160.144.191	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
95	4.583546666	34.160.144.191	172.16.96.4	TLSv1.2	434	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message, Application Data
100	4.583546981	34.160.144.191	172.16.96.4	TLSv1.2	434	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message, Application Data
101	4.583494448	34.160.144.191	172.16.96.4	TLSv1.2	1468	Server Hello
103	4.583546666	34.160.144.191	172.16.96.4	TLSv1.2	1494	Certificate
107	4.583537794	34.160.144.191	172.16.96.4	TLSv1.2	222	Server Key Exchange, Server Hello Done
108	4.583546801	172.16.96.4	34.160.144.191	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
111	4.573901777	34.160.144.191	172.16.96.4	TLSv1.2	434	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message, Application Data
114	4.573660599	172.16.96.4	34.160.144.191	TLSv1.2	153	Application Data
115	4.577877581	172.16.96.4	34.160.144.191	TLSv1.2	178	Application Data
116	4.577907577	172.16.96.4	34.160.144.191	TLSv1.2	92	Application Data
118	4.578480138	172.16.96.4	34.160.144.191	TLSv1.2	85	Encrypted Alert
123	4.580338078	172.16.96.4	34.160.144.191	TLSv1.2	178	Application Data
124	4.581071190	172.16.96.4	34.160.144.191	TLSv1.2	85	Encrypted Alert
126	4.595601273	34.160.144.191	172.16.96.4	TLSv1.2	82	Application Data
127	4.595601273	34.160.144.191	172.16.96.4	TLSv1.2	82	Application Data

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In summary, no TLS packets appear at first because the VPN hides them inside encrypted OpenVPN data, but TLS records become visible when the browser performs an HTTPS request and the filter is applied on the interface carrying the TLS traffic directly.

1.3 Study the TLS/SSL records in detail:

No.	Time	Source	Destination	Protocol	Length	Info
18	14.90533555	172.16.96.4	34.109.144.191	TLSv1.2	274	Client Hello (SHA256WithRSAEncryption)
19	14.90643732	172.16.96.4	34.109.144.191	TLSv1.2	274	Client Hello (SHA256WithRSAEncryption)
20	14.92954888	34.109.144.191	172.16.96.4	TLSv1.2	1488	Server Hello
21	14.92954888	34.109.144.191	172.16.96.4	TLSv1.2	1488	Server Hello
22	14.929549726	34.109.144.191	172.16.96.4	TLSv1.2	1494	Certificate
23	14.929549753	34.109.144.191	172.16.96.4	TLSv1.2	1494	Certificate
28	14.931988253	34.109.144.191	172.16.96.4	TLSv1.2	222	Server Key Exchange, Server Hello Done
29	14.931988489	34.109.144.191	172.16.96.4	TLSv1.2	222	Server Key Exchange, Server Hello Done
30	14.934848509	172.16.96.4	34.109.144.191	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
34	14.935489230	172.16.96.4	34.109.144.191	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
36	14.952895673	34.109.144.191	172.16.96.4	TLSv1.2	434	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message, Application Data
37	14.953420085	34.109.144.191	172.16.96.4	TLSv1.2	434	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message, Application Data

The captured TLS/SSL records show the complete sequence of messages exchanged during the TLS 1.2 handshake between the client and the server. The handshake begins with the **Client Hello**, where the client indicates supported cipher suites and includes the Server Name Indication, revealing the intended host. The server responds with a **Server Hello**, selecting the protocol parameters, followed by the **Certificate** message containing the server's public key.

Subsequently, the **Server Key Exchange** and **Server Hello Done** messages finalize the server's part of the negotiation. The client then sends its **Client Key Exchange**, followed by the Change Cipher Spec and Encrypted Handshake Message, indicating that all further communication is now encrypted. Finally, the server sends a New Session Ticket and its own Change Cipher Spec and encrypted handshake confirmation.

After the handshake completes, the capture shows Application Data packets, which represent encrypted HTTP traffic exchanged over the established TLS session. The contents of these packets cannot be read, demonstrating the confidentiality provided by TLS.

1.4 Is TLS and SSL the same protocol?

No, TLS and SSL are not the same protocol. **TLS is the newer, improved, and more secure version of SSL.** Although TLS was originally based on SSL, the SSL versions are now considered outdated and insecure, while TLS is the current standard for encrypted communication.

1.5 What is a TLS/SSL handshake?

A TLS/SSL handshake is the initial negotiation process between a client and a server that establishes a secure connection. During the handshake, both parties agree on the encryption algorithms, exchange keys, verify the server's identity, and create the session keys that will be used to encrypt the data.

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1.6 What does the TLS/SSL protocol provide, give examples of its applications (at least 2)?

TLS/SSL provides **confidentiality, integrity, and authentication** for data transmitted over a network. It ensures that the communication is encrypted, cannot be modified, and that the client can verify the identity of the server.

Common applications include:

- HTTPS for secure web browsing.
- Secure email protocols such as SMTPS, IMAPS, or POP3S.
- VPNs and other secure remote access technologies.

1.7 Which versions of the protocol are currently the most popular?

The most commonly used and widely supported versions today are **TLS 1.2 and TLS 1.3**. TLS 1.3 is the newest and most secure version, but TLS 1.2 is still widely used for compatibility with older systems.

1.8 Which version offers higher security and why?

TLS 1.3 offers higher security because it removes outdated and vulnerable cryptographic algorithms, reduces the number of handshake messages, encrypts more parts of the negotiation, and improves protection against attacks such as downgrade and replay attacks.

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1.9 Which version offers higher performance?

TLS 1.3 offers higher performance because it uses a faster handshake with fewer round-trips and more efficient cryptographic algorithms. This reduces latency and speeds up the establishment of secure connections compared to TLS 1.2.

Based on the traffic captured, analyze the recorded TLS/SSL traffic:

1.10 Which version of the protocol has been captured?

The captured traffic shows TLS 1.2 communication. This is visible in the Wireshark records, where several packets are labeled as TLSv1.2, including messages such as Server Hello, Certificate, Server Key Exchange, Change Cipher Spec, and Application Data. These are characteristic of a TLS 1.2 handshake.

If the connection were using TLS 1.3, the handshake would look very different. TLS 1.3 only exposes the Client Hello and Server Hello messages, while all other handshake steps (key exchange, authentication, and session establishment) appear as “Encrypted Handshake Message”. TLS 1.3 does not show Server Key Exchange or Change Cipher Spec, as these elements were removed from the protocol.

1.11 What kind of messages does a handshake consist of (connection establishment)?

A TLS 1.2 handshake consists of several messages exchanged between the client and the server to negotiate security parameters and establish an encrypted session. In the captured traffic, the handshake includes:

- **Client Hello:** the client proposes supported TLS versions and cipher suites.
- **Server Hello:** the server selects the protocol parameters.
- **Certificate:** the server sends its certificate for authentication.
- **Server Key Exchange:** the server provides key exchange information.
- **Server Hello Done:** indicates the server has finished its part of the negotiation.
- **Client Key Exchange:** the client sends key material to establish the session key.
- **Change Cipher Spec:** the client and server switch to encrypted communication.
- **Finished / Encrypted Handshake Message:** confirms that the handshake is complete.

These steps complete the establishment of a secure TLS session.

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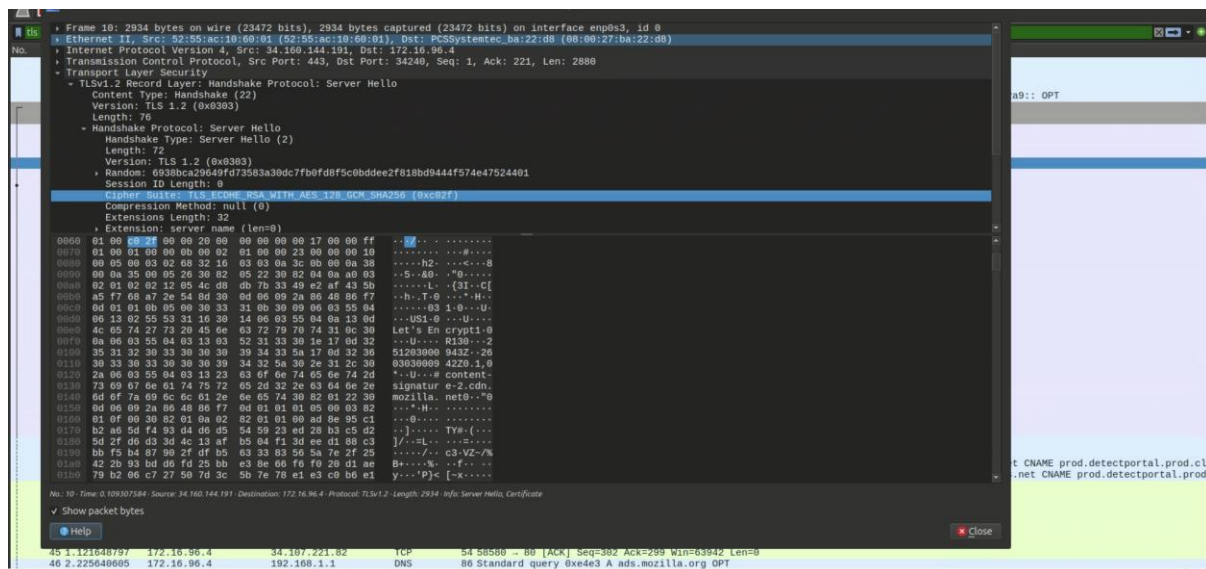
1.12 What does the version of the protocol use depend on?

The version of the TLS protocol used depends on several factors:

- **The client's supported TLS versions** (listed in the Client Hello).
- **The server's supported versions**, which determine the highest version both sides can use.
- **Security policies or configuration settings** on either side, which may disable older or insecure versions.
- **The cipher suites available**, since some are only compatible with specific TLS versions.

The final protocol version is the result of the negotiation between client and server during the handshake.

Take a print screen of what type of encryption was used for encryption and put it into the report.

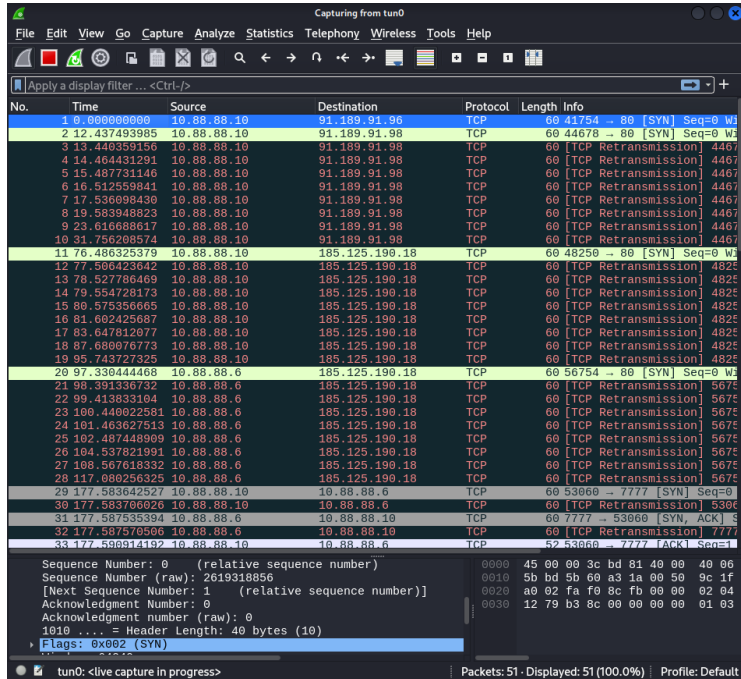


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2 OpenVPN Part-1:

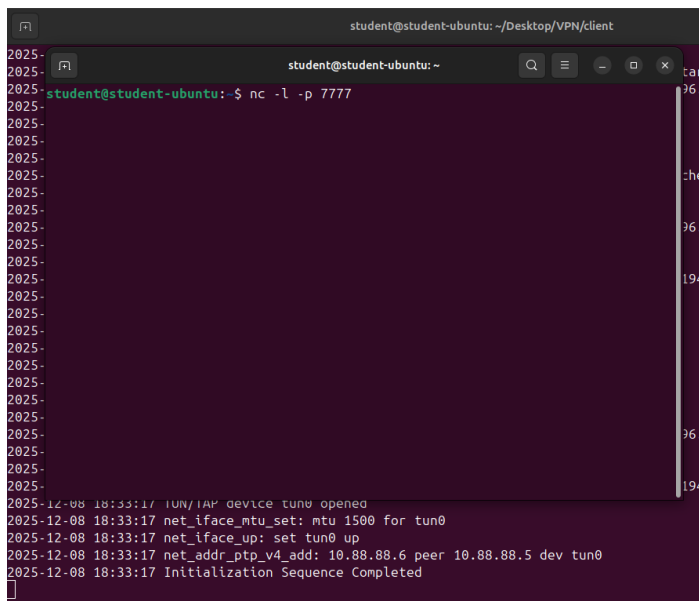
2.1 Capture the network traffic on the VPN network interface (usually, the name is tun0, check it using ifconfig command):

In Kali (Server) we start the connection (tun0):

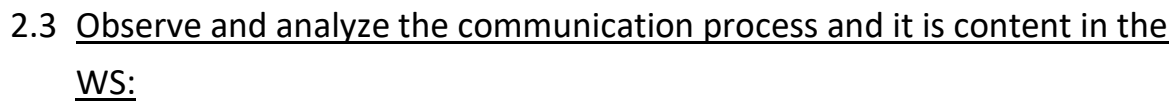


2.2 Send a few messages using netcat between clients A and B:

Client B: > nc -l -p 7777



And write some text in any of both terminals:



6	8.31840513	10.88.88.6	81.189.10.1	TCP	60 [TCP Retransmission] 50189 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240346936 TSecr=1130510672	WS-120
7	8.759374958	10.88.88.10	125.125.190.48	TCP	60 48334 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240344359 TSecr=WS-128	
8	0.743420728	10.88.88.10	185.125.190.48	TCP	60 [TCP Retransmission] 48334 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240344347 TSecr=WS-128	
9	0.743420728	10.88.88.10	185.125.190.48	TCP	60 [TCP Retransmission] 48334 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240344347 TSecr=WS-128	
10	0.774075298	10.88.88.10	185.125.190.48	TCP	60 [TCP Retransmission] 48334 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240344347 TSecr=WS-128	
11	0.774075298	10.88.88.10	185.125.190.48	TCP	60 54676 - 7777 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=1181015067 TSecr=WS-120	
12	0.774075298	10.88.88.10	185.125.190.48	TCP	60 54676 - 7777 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=1181015067 TSecr=WS-120	
13	1.78205453	10.88.88.6	10.88.88.10	TCP	60 7777 - 54676 [FIN, ACK] Seq=6 Ack=1 Win=61508 Len=0 MSS=1136 SACK_PERM Tsva1=328890162 TSecr=1181015067	WS-128
14	0.743420728	10.88.88.6	185.125.190.48	TCP	60 [TCP Retransmission] 7777 - 54676 [FIN, ACK] Seq=6 Ack=1 Win=61508 Len=0 MSS=1136 SACK_PERM Tsva1=328890162 TSecr=1181015067	
15	0.743420728	10.88.88.6	185.125.190.48	TCP	60 [TCP Retransmission] 7777 - 54676 [FIN, ACK] Seq=6 Ack=1 Win=61508 Len=0 MSS=1136 SACK_PERM Tsva1=328890162 TSecr=1181015067	
16	0.768347869	10.88.88.10	185.125.190.48	TCP	52 [TCP Dup ACK 15#1] 54676 - 7777 [ACK] Seq=6 Ack=1 Win=64256 Len=0 Tsva1=1181015067 TSecr=3288900173	
17	0.759045488	10.88.88.10	185.125.190.48	TCP	60 [TCP Retransmission] 48334 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240344347 TSecr=WS-128	
18	0.759045488	10.88.88.10	185.125.190.48	TCP	60 [TCP Retransmission] 48334 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240344347 TSecr=WS-128	
19	10.843954906	10.88.88.10	185.125.190.48	TCP	60 [TCP Retransmission] 48334 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240344040 TSecr=WS-128	
20	0.155452169	10.88.88.10	91.189.10.1	TCP	60 [TCP Retransmission] 48334 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240344040 TSecr=WS-128	
21	12.889832082	10.88.88.10	185.125.190.48	TCP	60 [TCP Retransmission] 48334 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240351514 TSecr=WS-128	
22	0.942755922	10.88.88.10	185.125.190.48	TCP	60 [TCP Retransmission] 48334 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240351514 TSecr=WS-128	
23	0.942755922	10.88.88.10	185.125.190.48	TCP	60 [TCP Retransmission] 48334 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240351514 TSecr=WS-128	
24	25.052355934	10.88.88.10	185.125.190.48	TCP	60 [TCP Retransmission] 48334 - 80 [SYN] Seq=6 Min=62420 Len=0 MSS=1136 SACK_PERM Tsva1=2240363674 TSecr=WS-128	
25	37.061299189	10.88.88.10	10.88.88.6	TCP	57 54676 - 7777 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=0 Tsva1=1181045037 TSecr=328890613	
26	0.155452169	10.88.88.10	91.189.10.1	TCP	57 54676 - 7777 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=0 Tsva1=1181045037 TSecr=328890613	
27	37.105904398	10.88.88.6	10.88.88.10	TCP	52 7777 - 54676 [ACK] Seq=1 Ack=6 Win=65280 Len=0 Tsva1=3288909052 TSecr=1181045037	
28	0.155452169	10.88.88.6	185.125.190.48	TCP	52 [TCP Dup ACK 27#1] 54676 - 7777 [ACK] Seq=6 Ack=1 Win=65280 Len=0 Tsva1=3288909052 TSecr=1181045037	
29	0.155452169	10.88.88.6	185.125.190.48	TCP	52 54676 - 7777 [ACK] Seq=6 Ack=1 Win=65280 Len=0 Tsva1=3288909052 TSecr=1181045037	
30	33.493535571	10.88.88.6	10.88.88.10	TCP	52 [TCP Retransmission] 7777 - 54676 [PSH, ACK] Seq=1 Ack=6 Win=65280 Len=0 Tsva1=3288917824 TSecr=1181045037	
31	44.96651510	10.88.88.10	10.88.88.6	TCP	53 54676 - 7777 [ACK] Seq=6 Ack=1 Win=64256 Len=0 Tsva1=1181035283 TSecr=3288917324	
32	0.155452169	10.88.88.10	91.189.10.1	TCP	52 54676 - 7777 [ACK] Seq=6 Ack=1 Win=65280 Len=0 Tsva1=3288917824 TSecr=1181045037	
33	47.499060571	10.88.88.6	10.88.88.10	TCP	52 7777 - 54676 [FIN, ACK] Seq=12 Ack=6 Win=65280 Len=0 Tsva1=328918969 TSecr=1181035283	
34	0.749511576	10.88.88.6	185.125.190.48	TCP	52 [TCP Retransmission] 7777 - 54676 [FIN, ACK] Seq=12 Ack=6 Win=65280 Len=0 Tsva1=328918969 TSecr=1181035283	
35	0.749511576	10.88.88.6	185.125.190.48	TCP	52 54676 - 7777 [ACK] Seq=6 Ack=1 Win=64256 Len=0 Tsva1=1181035283 TSecr=3288918969	
36	47.499060571	10.88.88.6	185.125.190.48	TCP	52 [TCP Dup ACK 35#1] 54676 - 7777 [ACK] Seq=6 Ack=1 Win=64256 Len=0 Tsva1=1181035283 TSecr=3288918969	
37	48.905287045	10.88.88.10	10.88.88.6	TCP	52 [TCP Retransmission] 54676 - 7777 [FIN, ACK] Seq=6 Ack=1 Win=64256 Len=0 Tsva1=1181035876 TSecr=328918969	

We can analyze in Wireshark given the known port 7777, apply the filter `> tcp.port == 7777`

The image displays a Wireshark packet capture of a TCP connection. The top toolbar shows standard network analysis tools. The packet list pane on the left shows a sequence of packets, with packet 11 selected. The packet details pane on the right shows the structure of the selected packet, including the Ethernet II header, Internet Protocol header, and Transmission Control Protocol header. The packet bytes pane at the bottom shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
11	7.788286195	10.88.88.10	10.88.88.6	TCP	60	54670 → 7777 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=1183015672 TSecr=0 WS=128
12	7.788286490	10.88.88.10	10.88.88.6	TCP	60	7777 → 54670 [SYN, ACK] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=1183015672 TSecr=0 WS=128
13	7.788286490	10.88.88.10	10.88.88.6	TCP	60	7777 → 54670 [SYN, ACK] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=1183015672 TSecr=0 WS=128
14	7.788288811	10.88.88.6	10.88.88.10	TCP	60	1183015672 → 7777 [ACK] Seq=65160 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3288880173 TSecr=1183015672 WS=128
15	7.785336228	10.88.88.10	10.88.88.6	TCP	52	54670 → 7777 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1183015677 TSecr=3288880173
16	7.785344780	10.88.88.10	10.88.88.6	TCP	52	7777 → 54670 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1183015677 TSecr=3288880173
17	7.761299189	10.88.88.10	10.88.88.6	TCP	52	54670 → 7777 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=5 TSval=1183045037 TSecr=3288880173
18	7.761343240	10.88.88.10	10.88.88.6	TCP	52	7777 → 54670 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1183045037 TSecr=3288880173
27	37.165994369	10.88.88.6	10.88.88.10	TCP	52	7777 → 54670 [ACK] Seq=1 Ack=6 Win=65280 Len=0 TSval=3288999542 TSecr=1183045037
28	37.165995629	10.88.88.6	10.88.88.10	TCP	52	7777 → 54670 [ACK] Seq=1 Ack=6 Win=65280 Len=0 TSval=3288999542 TSecr=1183045037
29	44.953864128	10.88.88.6	10.88.88.10	TCP	63	1183045037 → 7777 [ACK] Seq=65160 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3288917324 TSecr=1183045037
30	44.953864128	10.88.88.6	10.88.88.10	TCP	63	1183045037 → 7777 [ACK] Seq=65160 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3288917324 TSecr=1183045037
31	44.966815116	10.88.88.10	10.88.88.6	TCP	52	54670 → 7777 [ACK] Seq=6 Ack=12 Win=64256 Len=0 TSval=1183052835 TSecr=3288917324
32	44.966815138	10.88.88.10	10.88.88.6	TCP	52	7777 → 54670 [FIN, ACK] Seq=12 Ack=6 Win=65280 Len=0 TSval=3288919869 TSecr=1183052835
33	47.498919672	10.88.88.6	10.88.88.10	TCP	52	7777 → 54670 [ACK] Seq=6 Ack=13 Win=64256 Len=0 TSval=1183052835 TSecr=3288919869
34	47.498919672	10.88.88.6	10.88.88.10	TCP	52	7777 → 54670 [ACK] Seq=6 Ack=13 Win=64256 Len=0 TSval=1183052835 TSecr=3288919869
35	47.545753616	10.88.88.10	10.88.88.6	TCP	52	54670 → 7777 [FIN, ACK] Seq=6 Ack=13 Win=64256 Len=0 TSval=1183052835 TSecr=3288919869
36	47.545753616	10.88.88.10	10.88.88.6	TCP	52	54670 → 7777 [FIN, ACK] Seq=6 Ack=13 Win=64256 Len=0 TSval=1183052835 TSecr=3288919869
37	49.052827045	10.88.88.10	10.88.88.6	TCP	52	7777 → 54670 [ACK] Seq=13 Ack=7 Win=65280 Len=0 TSval=3288919869 TSecr=1183052835
38	49.052827045	10.88.88.10	10.88.88.6	TCP	52	7777 → 54670 [ACK] Seq=13 Ack=7 Win=65280 Len=0 TSval=3288919869 TSecr=1183052835
39	49.052827045	10.88.88.10	10.88.88.6	TCP	52	7777 → 54670 [ACK] Seq=13 Ack=7 Win=65280 Len=0 TSval=3288919869 TSecr=1183052835
40	49.052827045	10.88.88.10	10.88.88.6	TCP	52	7777 → 54670 [ACK] Seq=13 Ack=7 Win=65280 Len=0 TSval=3288919869 TSecr=1183052835

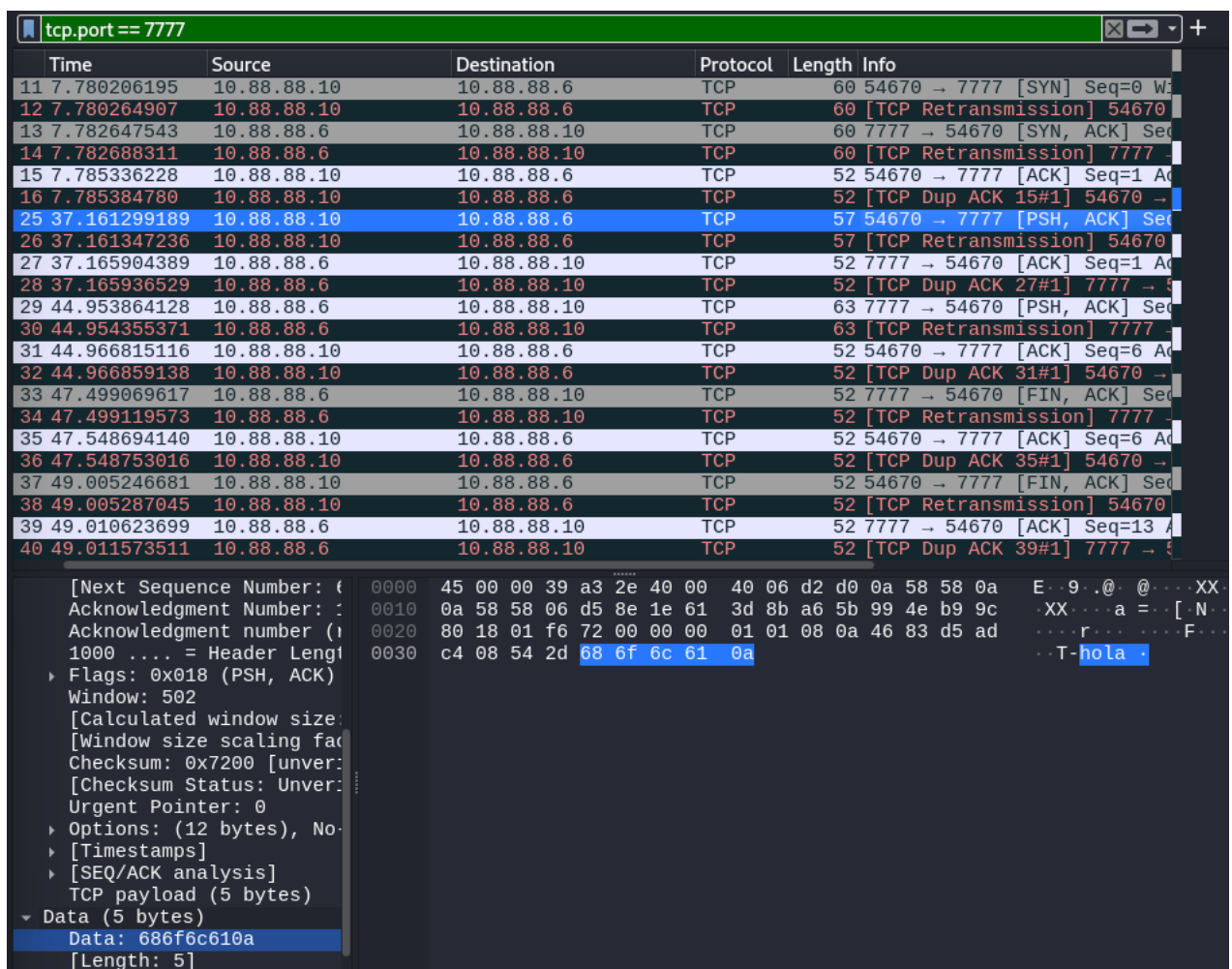
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I could see TCP packets between the VPN addresses 10.88.88.6 (client A – Ubuntu1) and 10.88.88.10 (client B – Ubuntu 2), using destination port 7777.

This confirms that the captured traffic corresponds to the netcat communication inside the VPN tunnel.

2.4 Find and verify the data sent by netcat in the WS. Is it possible to read the message that was sent by netcat? Take a print screen of the data contained in the packets:

By selecting one of the TCP packets (EX: 25) with payload (Flag [PSH, ACK]) and expanding the *Data* section in Wireshark, you can see the message sent via netcat ("hola").



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Also, the second message is in frame 29 so we can see its content (“Im ClientB”):

The image shows a Wireshark packet capture of a TCP connection. The top pane displays a list of packets, with frame 29 selected. The packet details pane shows the structure of the selected packet, including the TCP header and the data field. The data field contains the text "Im ClientB".

Time	Source	Destination	Protocol	Length	Info
12 7.780264907	10.88.88.10	10.88.88.6	TCP	60	[TCP Retransmission] 54670 → 7777 [ACK, Seq=154670]
13 7.782647543	10.88.88.6	10.88.88.10	TCP	60	7777 → 54670 [SYN, ACK] Seq=154670
14 7.782688311	10.88.88.6	10.88.88.10	TCP	60	[TCP Retransmission] 7777 → 54670 [ACK, Seq=154670]
15 7.785336228	10.88.88.10	10.88.88.6	TCP	52	54670 → 7777 [ACK] Seq=154670
16 7.785384780	10.88.88.10	10.88.88.6	TCP	52	[TCP Dup ACK 15#1] 54670 → 7777
25 37.161299189	10.88.88.10	10.88.88.6	TCP	57	54670 → 7777 [PSH, ACK] Seq=154670
26 37.161347236	10.88.88.10	10.88.88.6	TCP	57	[TCP Retransmission] 54670 → 7777 [PSH, ACK] Seq=154670
27 37.165904389	10.88.88.6	10.88.88.10	TCP	52	7777 → 54670 [ACK] Seq=154670
28 37.165936529	10.88.88.6	10.88.88.10	TCP	52	[TCP Dup ACK 27#1] 7777 → 54670 [ACK] Seq=154670
29 44.953864128	10.88.88.6	10.88.88.10	TCP	63	7777 → 54670 [PSH, ACK] Seq=154670
30 44.954355371	10.88.88.6	10.88.88.10	TCP	63	[TCP Retransmission] 7777 → 54670 [PSH, ACK] Seq=154670
31 44.966815116	10.88.88.10	10.88.88.6	TCP	52	54670 → 7777 [ACK] Seq=154670
32 44.966859138	10.88.88.10	10.88.88.6	TCP	52	[TCP Dup ACK 31#1] 54670 → 7777
33 47.499069617	10.88.88.6	10.88.88.10	TCP	52	7777 → 54670 [FIN, ACK] Seq=154670
34 47.499119573	10.88.88.6	10.88.88.10	TCP	52	[TCP Retransmission] 7777 → 54670 [FIN, ACK] Seq=154670
35 47.548694140	10.88.88.10	10.88.88.6	TCP	52	54670 → 7777 [ACK] Seq=154670
36 47.548753016	10.88.88.10	10.88.88.6	TCP	52	[TCP Dup ACK 35#1] 54670 → 7777
37 49.005246681	10.88.88.10	10.88.88.6	TCP	52	54670 → 7777 [FIN, ACK] Seq=154670
38 49.005287045	10.88.88.10	10.88.88.6	TCP	52	[TCP Retransmission] 54670 → 7777 [FIN, ACK] Seq=154670
39 49.010623699	10.88.88.6	10.88.88.10	TCP	52	7777 → 54670 [ACK] Seq=154670
40 49.011573511	10.88.88.6	10.88.88.10	TCP	52	[TCP Dup ACK 39#1] 7777 → 54670 [ACK] Seq=154670

Packet 29 details:

- Next Sequence Number: 7777
- Acknowledgment Number: 154670
- Acknowledgment number (offset): 1000
- Flags: 0x018 (PSH, ACK)
- Window: 510
- Checksum: 0x05d4
- Urgent Pointer: 0
- Options: (12 bytes), No-ops, [Timestamps], [SEQ/ACK analysis]
- TCP payload (11 bytes)
- Data (11 bytes): 496d20436c696556e74

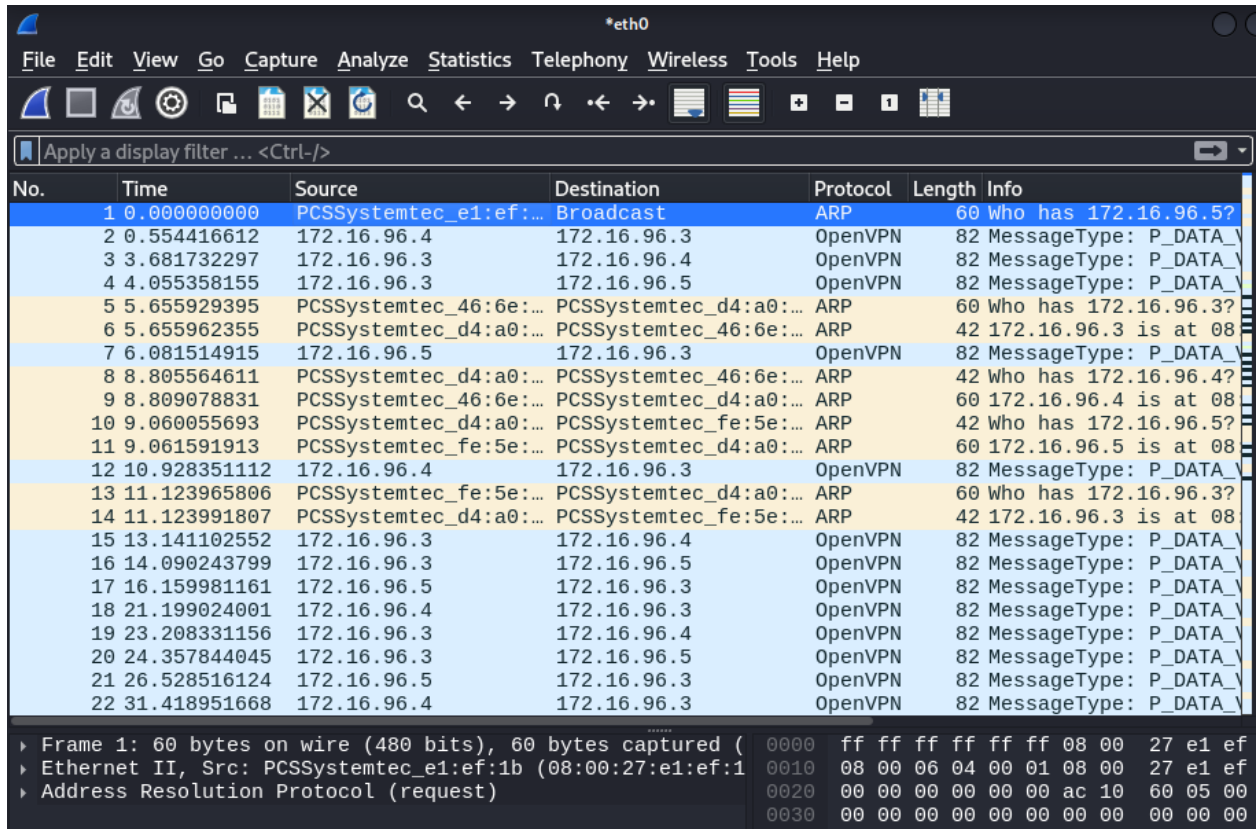
Therefore, it is possible to read the message sent by netcat as they are not cyphered at this point when capturing on the VPN interface tun0, because the traffic is already decrypted through the server network.

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3 OpenVPN Part-2:

3.1 Capture the network traffic on the physical network interface (usually, the name starts with en...):

In Kali (Server) we start the connection (eth0):



The image shows a Wireshark packet capture on the eth0 interface. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	PCSSystemtec_e1:ef:...	Broadcast	ARP	60	Who has 172.16.96.5?
2	0.554416612	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V
3	3.681732297	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V
4	4.055358155	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_V
5	5.655929395	PCSSystemtec_46:6e:...	PCSSystemtec_d4:a0:...	ARP	60	Who has 172.16.96.3?
6	5.655962355	PCSSystemtec_d4:a0:...	PCSSystemtec_46:6e:...	ARP	42	172.16.96.3 is at 08
7	6.081514915	172.16.96.5	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V
8	8.805564611	PCSSystemtec_d4:a0:...	PCSSystemtec_46:6e:...	ARP	42	Who has 172.16.96.4?
9	8.809078831	PCSSystemtec_46:6e:...	PCSSystemtec_d4:a0:...	ARP	60	172.16.96.4 is at 08
10	9.060055693	PCSSystemtec_d4:a0:...	PCSSystemtec_fe:5e:...	ARP	42	Who has 172.16.96.5?
11	9.061591913	PCSSystemtec_fe:5e:...	PCSSystemtec_d4:a0:...	ARP	60	172.16.96.5 is at 08
12	10.928351112	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V
13	11.123965806	PCSSystemtec_fe:5e:...	PCSSystemtec_d4:a0:...	ARP	60	Who has 172.16.96.3?
14	11.123991807	PCSSystemtec_d4:a0:...	PCSSystemtec_fe:5e:...	ARP	42	172.16.96.3 is at 08
15	13.141102552	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V
16	14.090243799	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_V
17	16.159981161	172.16.96.5	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V
18	21.199024001	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V
19	23.208331156	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V
20	24.357844045	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_V
21	26.528516124	172.16.96.5	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V
22	31.418951668	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V

Below the table, the details pane shows the selected packet (Frame 1):

```

Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (
Ethernet II, Src: PCSSystemtec_e1:ef:1b (08:00:27:e1:ef:1
Address Resolution Protocol (request)
  
```

3.2 Send a few messages using netcat between clients A and B:

```

student@student-ubuntu:~$ nc -l -p 7777
ClientA calling ClientB sure?
Yea all connections are clear

student@student-ubuntu:~$ nc 10.88.88.6 7777
ClientA calling ClientB sure?
Yea all connections are clear
  
```

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3.3 Observe and analyze the communication process and its content in the WS:

We apply the filter to only register messages of Client A (172.16.96.4) and Client B (172.16.96.5):

ip.addr == 172.16.96.4 ip.addr == 172.16.96.5						
No.	Time	Source	Destination	Protocol	Length	Info
2	0.554416612	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_
3	3.681732297	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_
4	4.055358155	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_
7	6.081514915	172.16.96.5	172.16.96.3	OpenVPN	82	MessageType: P_DATA_
12	10.928351112	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_
15	13.141102552	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_
16	14.090243799	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_
17	16.159981161	172.16.96.5	172.16.96.3	OpenVPN	82	MessageType: P_DATA_
18	21.199024001	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_
19	23.208331156	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_
20	24.357844045	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_
21	26.528516124	172.16.96.5	172.16.96.3	OpenVPN	82	MessageType: P_DATA_
22	31.418951668	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_
23	33.260508644	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_
24	34.619129808	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_
25	36.761655748	172.16.96.5	172.16.96.3	OpenVPN	82	MessageType: P_DATA_
26	41.503910619	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_
29	42.484163147	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_
31	43.438391544	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_
32	43.524740848	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_
34	44.549941868	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_
35	44.552152574	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_

As we can see, all packages pass through the VPN server (172.16.96.3).

3.4 Find and verify the data sent by netcat in the WS. Take a print screen of the data contained in the packets:

31	43.438391544	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_
32	43.524740848	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_
34	44.549941868	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_
35	44.552152574	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_

Frame 31: 82 bytes on wire (656 bits) captured (82 bytes) on interface eth0	0000	08 00 27 46 6e 10 08 00 27 d4 a0 ae 08 00 45 00	E
Ethernet II, Src: PCSyste	0010	00 44 16 9e 40 00 40 11 0b e3 ac 10 60 03 ac 10	..D..@..jH...
Internet Protocol Version	0020	60 04 04 aa a4 34 00 30 18 6a 48 00 00 01 00 00	...4.0.jH...
User Datagram Protocol, Sr	0030	00 9a f7 44 9a 1a c2 47 35 45 72 8a e3 4d b7 60	...D...G 5Er..M..
Source Port: 1194	0040	6f 7f fc 5a 22 d8 e7 a3 75 c1 72 9e 1a 6a 81 44	o..Z"...u.r..j.D
Destination Port: 42036	0050	48 7c	H
Length: 48			
Checksum: 0x186a [unveri			
[Checksum Status: Unveri			
[Stream index: 0]			
[Timestamps]			
UDP payload (40 bytes)			
OpenVPN Protocol			
Type: 0x48 [opcode/key_i			
Peer ID: 1			
Data (36 bytes)			
Data: 0000009af7449a1			

As the data is encrypted through the network thanks to the VPN, no critical data is shown in Wireshark.

So, through the eth0 / enp0s3 network that is the 172.16.96.0/24 network the data is encrypted.

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Questions to task 2 and 3:

3.5 Is it possible to read the message that was sent via netcat?

- **tun0 (Task 2):** Yes, as the tunnel terminates and decrypts the traffic, so Wireshark captures TCP between 10.88.88.6 and 10.88.88.10 on port 7777 and the netcat message is visible as plain text in the Data field.
- **enp0s3 (Task 3):** No, as the packets are OpenVPN UDP/1194 between 10.0.2.3 and 10.0.2.5 and their payloads are encrypted, so the netcat message cannot be read.

3.6 What is the characteristic of captured traffic on the VPN network interface and the host interface?

VPN interface (tun0):

- Source and destination IP addresses correspond to the VPN-assigned addresses (10.88.88.x).
- The protocol appears as the actual application protocol (e.g., **TCP on port 7777 for netcat**).
- The **payload is visible in clear text** because the traffic is already decrypted at this point.

Host/physical interface (enp0s3 / eth0):

- Source and destination IP addresses **are the real network addresses of client A and server (10.0.2.4 <-> 10.0.2.3) or client B and server (10.0.2.5 <-> 10.0.2.3)**.
- The observed protocol is OpenVPN encapsulated in **UDP source port 1194**.
- The **payload is fully encrypted**, so no application-layer data is visible.

3.7 What relevant information can you see analyzing the traffic for both cases?

From the physical interface (enp0s3 / eth0) you observe:

- The existence of a VPN tunnel between the client and the server (traffic between 10.0.2.3:xxxx and 10.0.2.5:1194).
- The **volume, direction, and timing** of the encrypted packets but **no visibility** into the application data since the **tunnel payload is encrypted**.

From the VPN interface (tun0) you observe:

- The **VPN-level source and destination addresses** of the communicating hosts.
- The **application protocol and port being used** (such as netcat on TCP 7777).
- The **full message contents**, which appear in **clear text** after decryption.

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4 OpenVPN Part-3:

4.1 Capture the network traffic on the physical network interface (usually, the name starts with en...):

In Kali (Server) we start the connection (eth0):

No.	Time	Source	Destination	Protocol	Length	Info
0	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
1	0.000000	172.16.96.5	172.16.96.3	OpenVPN	128	MessageType: P_DATA_V2
2	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
3	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
4	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
5	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
6	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
7	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
8	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
9	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
10	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
11	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
12	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
13	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
14	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
15	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
16	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
17	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
18	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
19	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
20	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
21	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
22	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
23	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
24	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
25	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
26	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
27	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
28	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
29	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
30	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
31	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
32	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
33	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
34	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
35	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
36	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
37	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
38	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
39	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128
40	0.000000	10.88.88.6	172.16.96.3	TCP	74	[TCP Retransmission] 39672 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3919768340 TSecr=0 WS=128

4.2 Send a larger file via netcat. The file is located on the client's VM. (in directory: /home/Ubuntu/Desktop/FILES/ubuntu.iso, ubuntu.iso ~900MB):

We need to write to send the file ("FileToSend.iso" of 627MB):

ClientB: nc -l -p 7777 > FileToReceive.iso

ClientA: cat ./Desktop/ FileToSend.iso | nc 10.88.88.6 7777 (Where 10.88.88.6 is the address of client B)

(As cat writes the data and the pipe inserts it into the connection that clientB receives and converts adds it into the file "FileToReceive.iso")

```
student@student-ubuntu:~$ nc -l -p 7777 > FileToReceive.iso
```

```
student@student-ubuntu:~$ cat ./Desktop/FileToSend.iso | nc 10.88.88.6 7777
```

To check if it is being sent check Wireshark:

CYBERSECURITY REPORT – LABORATORY 9

No.	Time	Source	Destination	Protocol	Length	Info
257352	111.023626141	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
257353	111.024081223	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257354	111.024652811	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257355	111.025811643	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
257356	111.026301572	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
257357	111.026886119	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
257358	111.027444417	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
257359	111.027974442	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
257360	111.028563368	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257361	111.028870335	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
257362	111.029353282	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257363	111.029353471	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257364	111.030132553	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257365	111.030132710	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257366	111.030698586	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257367	111.031200790	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
257368	111.031400542	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
257369	111.031893039	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
257370	111.032398368	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
257371	111.032932006	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
257372	111.033249693	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257373	111.033763150	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257374	111.033763326	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
257375	111.033763385	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257376	111.033763417	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257377	111.033979882	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
257378	111.034329105	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
257379	111.034636706	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
257380	111.035376986	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
257381	111.035377158	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
257382	111.035554949	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
257383	111.035878519	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2

Thousand packages are sent/received so it is working.

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4.3 Transfer the file 4 times, each time change the encryption and authentication algorithm. Edit client.ovpn for both clients and change the following parameters simultaneously:

By opening Wireshark and checking the frame times we can calculate the duration of the transaction.

4.3.1 Setting 1:

- Cipher AES-256-GCM
- Auth SHA512,

Packets of 1242 bytes and 118 bytes:

The large ones (1242 B) are the encrypted data itself, and the 118 packages are the confirmations that the data arrived

First Frame Time: 2.48 seconds

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	PCSystemtec_e1:ef...	Broadcast	ARP	60	Who has 172.16.96.4? Tell 172.16.96.2
2	0.115432356	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
3	0.115853275	10.88.88.10	91.189.91.48	TCP	74	50094 - 80 [EST] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3101088564 TSecr=0 WS=128
4	2.163456370	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_V2
5	2.477835526	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
6	2.478332567	172.16.96.3	172.16.96.5	OpenVPN	126	MessageType: P_DATA_V2
7	2.483583924	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
8	2.484306687	172.16.96.3	172.16.96.4	OpenVPN	126	MessageType: P_DATA_V2
9	2.485293171	172.16.96.4	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
10	2.489726587	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
11	2.489726961	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
12	2.489727113	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
13	2.489727242	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
14	2.489727372	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
15	2.489727504	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
16	2.489727639	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
17	2.489727764	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
18	2.489784295	172.16.96.3	172.16.96.5	OpenVPN	118	MessageType: P_DATA_V2
19	2.490103879	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
20	2.490384228	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
21	2.490484786	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
22	2.492419536	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
23	2.493391039	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
24	2.493730456	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
25	2.493817466	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
26	2.509687597	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
27	2.509688280	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
28	2.509688346	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
29	2.511445343	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
30	2.511562690	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
31	2.511602726	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
32	2.511861150	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
33	2.512474868	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
34	2.515082455	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2

Last Frame Time: 875.4 seconds

No.	Time	Source	Destination	Protocol	Length	Info
1535749	875.393190400	172.16.96.3	172.16.96.5	OpenVPN	274	MessageType: P_DATA_V2
1535750	875.393507159	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535751	875.394241050	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1535752	875.394368721	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535753	875.394706191	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535754	875.394859926	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1535755	875.395099769	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535756	875.395576360	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535757	875.396020943	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535758	875.396557536	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535759	875.397121067	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535760	875.397687185	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535761	875.398379983	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535762	875.398774661	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535763	875.399094931	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535764	875.399420910	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535765	875.399743314	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535766	875.400062924	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535767	875.400370990	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535768	875.400733272	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535769	875.401068049	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535770	875.401381726	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535771	875.401692048	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535772	875.402100952	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535773	875.402498246	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535774	875.402812089	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535775	875.403133552	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1535776	877.815415344	PCSystemtec_fe:5a...	PCSystemtec_d4:a0...	ARP	60	Who has 172.16.96.3? Tell 172.16.96.5

Time it took to move the file: 872.92 seconds

No application-layer data is readable on this interface.

CYBERSECURITY REPORT – LABORATORY 9

4.3.2 Setting 2:

- Cipher AES-128-CBC
- Auth SHA1

First Frame Time: 3.29 seconds

7	3.084974785	PCSSystemtec_fe:5e...	PCSSystemtec_d4:a0...	ARP	60	172.16.96.5 is at 08:00:27:fe:5e:16
9	3.278830239	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
9	3.280978299	172.16.96.3	172.16.96.5	OpenVPN	126	MessageType: P_DATA_V2
10	3.283570724	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
11	3.285965688	172.16.96.3	172.16.96.4	OpenVPN	126	MessageType: P_DATA_V2
12	3.287978847	172.16.96.4	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
13	3.288764740	172.16.96.3	172.16.96.5	OpenVPN	118	MessageType: P_DATA_V2
14	3.290493133	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
15	3.290892430	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
16	3.290892430	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
17	3.290893223	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
18	3.290893325	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
19	3.290893495	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
20	3.290893580	172.16.96.5	172.16.96.3	OpenVPN	119	MessageType: P_DATA_V2
21	3.290893606	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
22	3.290896473	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
23	3.290896659	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
24	3.290896679	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
25	3.291935804	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
26	3.291411835	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
27	3.291723937	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
28	3.292090973	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
29	3.292138467	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
30	3.292392953	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
31	3.293822580	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
32	3.293840838	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
33	3.296014890	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2

Last Frame Time: 614.19 seconds

No.	Time	Source	Destination	Protocol	Length	Info
1584529	614.173850871	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1584530	614.174405085	172.16.96.3	172.16.96.5	OpenVPN	118	MessageType: P_DATA_V2
1584531	614.174934637	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584532	614.175493931	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584533	614.176078260	172.16.96.4	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1584534	614.176636513	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584535	614.178452890	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584536	614.179193101	172.16.96.4	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584537	614.179733224	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584538	614.180364656	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584539	614.181159741	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584540	614.181631088	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584541	614.182091516	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584542	614.182571296	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584543	614.183180781	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584544	614.183721115	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584545	614.184333562	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584546	614.185227135	172.16.96.4	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584547	614.185781767	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584548	614.186372610	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584549	614.186942181	172.16.96.4	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584550	614.187513911	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584551	614.188123217	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584552	614.189016108	172.16.96.4	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584553	614.189545766	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584554	614.190379920	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1584555	624.262363533	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V2
1584556	624.264892915	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V2
1584557	624.441587198	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_V2
1584558	624.440686012	172.16.96.5	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V2
1584559	629.512813477	PCSSystemtec_d4:a0...	PCSSystemtec_fe:5e...	ARP	42	Who has 172.16.96.5? Tell 172.16.96.3
1584560	629.514737078	PCSSystemtec_fe:5e...	PCSSystemtec_d4:a0...	ARP	60	172.16.96.5 is at 08:00:27:fe:5e:16
1584561	633.613394546	172.16.96.5	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V2
1584562	634.633755435	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V2
1584563	634.637183381	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V2
1584564	634.926482352	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_V2

Time it took to move the file: 610.9 seconds

4.3.3 Setting 3:

- Cipher DES-EDE-CBC
- Auth MD5

First Frame Time: 6.69 seconds

13	4.374275243	10.66.66.0	91.169.91.49	TCP	74	ICP (Transmission) 50904 - 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=2710993514 TSecr=0 WS=128
14	5.400254924	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
15	5.400463570	172.16.96.3	172.16.96.5	TCP	74	ICP (Transmission) 50904 - 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=2710993539 TSecr=0 WS=128
16	6.039025411	172.16.96.5	172.16.96.3	OpenVPN	114	MessageType: P_DATA_V2
17	6.079935422	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
18	6.080706545	172.16.96.3	172.16.96.5	OpenVPN	126	MessageType: P_DATA_V2
19	6.083556640	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
20	6.085255981	172.16.96.3	172.16.96.4	OpenVPN	126	MessageType: P_DATA_V2
21	6.087925551	172.16.96.4	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
22	6.088829842	172.16.96.3	172.16.96.5	OpenVPN	118	MessageType: P_DATA_V2
23	6.090331120	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
24	6.091771701	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
25	6.090166474	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
26	6.090464951	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
27	6.090723062	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
28	6.091330710	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
29	6.091608252	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
30	6.091608625	172.16.96.5	172.16.96.3	OpenVPN	119	MessageType: P_DATA_V2
31	6.091601061	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
32	6.092194978	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
33	6.092317170	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
34	6.092388675	172.16.96.3	172.16.96.4	OpenVPN	119	MessageType: P_DATA_V2
35	6.092707193	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
36	6.092861613	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
37	6.092963378	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
38	6.093107650	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
39	6.094206750	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
40	6.094208079	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
41	6.094209187	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
42	6.094569678	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2

Last Frame Time: 617.92 seconds

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No.	Time	Source	Destination	Protocol	Length	Info
1601595	617.909735585	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601596	617.909786627	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601597	617.911899065	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601598	617.911899513	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601599	617.911899636	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601600	617.911899669	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601601	617.911899697	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601602	617.912380278	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601603	617.912779352	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601604	617.913319776	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601605	617.913539269	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601606	617.914091178	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601607	617.914519886	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601608	617.914605844	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601609	617.915181756	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601610	617.915537655	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601611	617.916098468	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601612	617.916476441	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601613	617.916711418	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601614	617.917119349	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601615	617.917521677	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601616	617.917881292	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601617	617.918478914	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601618	617.919216826	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601619	617.919758537	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601620	617.920142809	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601621	617.920492384	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601622	617.920851602	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601623	617.921231174	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601624	617.921617871	172.16.96.4	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601625	617.921984687	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601626	617.922414705	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1601627	617.922950344	172.16.96.4	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1601628	627.992598841	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_V2
1601629	627.995158405	172.16.96.5	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V2
1601630	628.309364906	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V2

Time it took to move the file: 611.23 seconds

4.3.4 Setting 4:

- Cipher DES-CBC (As it is not supported, I changed it to BF-CBC)
- Auth MD5

First Frame Time: 5.24 seconds

No.	Time	Source	Destination	Protocol	Length	Info
1	0.009000000	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
2	0.003971774	10.88.88.6	185.125.190.17	TCP	74	56600 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3372248255 TSecr=0 WS=128
3	0.199026057	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
4	0.1990690284	10.88.88.10	185.125.190.18	TCP	74	36182 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=1555148359 TSecr=0 WS=128
5	0.779086290	PCSSystemtec_d4:a0::	52:55:ac:10:60:01	ARP	42	Who has 172.16.96.17 Tell 172.16.96.3
6	0.771262288	52:55:ac:10:60:01	PCSSystemtec_d4:a0::	ARP	64	172.16.96.1 is at 52:55:ac:10:60:01
7	0.824604255	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
8	1.025314801	10.88.88.6	185.125.190.17	TCP	74	[TCP Retransmission] 56600 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3372241279 TSecr=0 WS=128
9	1.795497535	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
10	1.819773460	10.88.88.10	91.189.91.96	TCP	74	34550 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=1589863856 TSecr=0 WS=128
11	1.242315314	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
12	2.242759723	10.88.88.10	185.125.190.18	TCP	74	36188 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=1555150482 TSecr=0 WS=128
13	0.072693508	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
14	0.072693508	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
15	5.232574453	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
16	5.23242123	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V2
17	5.234263140	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
18	5.237388026	172.16.96.5	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
19	5.237832215	172.16.96.3	172.16.96.4	OpenVPN	126	MessageType: P_DATA_V2
20	5.239760807	172.16.96.4	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
21	5.240636793	172.16.96.3	172.16.96.5	OpenVPN	118	MessageType: P_DATA_V2
22	5.242424070	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
23	5.242434569	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
24	5.243363381	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
25	5.243514099	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
26	5.243907967	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
27	5.244459567	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
28	5.245166902	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
29	5.245167665	172.16.96.3	172.16.96.5	OpenVPN	1242	MessageType: P_DATA_V2
30	5.246222516	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
31	5.246526404	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
32	5.246766139	172.16.96.5	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
33	5.247142516	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
34	5.247142771	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
35	5.247615947	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2
36	5.247962482	172.16.96.4	172.16.96.3	OpenVPN	1242	MessageType: P_DATA_V2

Last Frame Time: 749.78 seconds

No.	Time	Source	Destination	Protocol	Length	Info
1633698	749.779421489	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1633699	749.779689742	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1633700	749.779819482	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1633701	749.780122835	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1633702	749.780766537	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1633703	749.780997550	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1633704	749.781251295	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1633705	749.781609080	172.16.96.4	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1633706	749.781951547	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1633707	749.782065548	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1633708	749.782769520	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1633709	749.782963598	172.16.96.4	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1633710	749.783721208	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1633711	749.783918293	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1633712	749.784180056	172.16.96.5	172.16.96.3	OpenVPN	118	MessageType: P_DATA_V2
1633713	749.784294553	172.16.96.3	172.16.96.4	OpenVPN	118	MessageType: P_DATA_V2
1633714	753.507254762	PCSSystemtec_d4:a0::	PCSSystemtec_d4:a0::	ARP	60	Who has 172.16.96.3? Tell 172.16.96.4
1633715	753.507271011	PCSSystemtec_d4:a0::	PCSSystemtec_d4:a0::	ARP	42	172.16.96.3 is at 08:00:27:d4:a0:a6
1633716	754.464165039	PCSSystemtec_d4:a0::	PCSSystemtec_d4:a0::	ARP	42	Who has 172.16.96.4? Tell 172.16.96.3
1633717	754.465634025	PCSSystemtec_d4:a0::	PCSSystemtec_d4:a0::	ARP	60	172.16.96.4 is at 08:00:27:46:6e:10
1633718	757.237541413	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V2
1633719	758.193671969	172.16.96.5	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V2
1633720	759.360698081	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V2
1633721	761.228293299	PCSSystemtec_e1:ef::	Broadcast	60	Who has 172.16.96.4? Tell 172.16.96.2	
1633722	762.139965177	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_V2
1633723	767.746152304	172.16.96.4	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V2
1633724	767.750243590	PCSSystemtec_d4:a0::	PCSSystemtec_fe:5e::	ARP	42	Who has 172.16.96.5? Tell 172.16.96.3
1633725	767.753570840	PCSSystemtec_fe:5e::	PCSSystemtec_d4:a0::	ARP	60	172.16.96.5 is at 08:00:27:fe:5e:16
1633726	768.214622215	172.16.96.5	172.16.96.3	OpenVPN	82	MessageType: P_DATA_V2
1633727	769.990645905	172.16.96.3	172.16.96.4	OpenVPN	82	MessageType: P_DATA_V2
1633728	772.280955625	172.16.96.3	172.16.96.5	OpenVPN	82	MessageType: P_DATA_V2
1633729	774.353496102	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
1633730	774.353909110	10.88.88.10	91.189.91.96	TCP	74	5530 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=3030689964 TSecr=0 WS=128
1633731	775.408421923	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2
1633732	775.408792648	10.88.88.6	91.189.91.96	TCP	74	[TCP Retransmission] 5530 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1136 SACK_PERM TSval=30306900110 TSecr=0 WS=128
1633733	776.430786578	172.16.96.4	172.16.96.3	OpenVPN	126	MessageType: P_DATA_V2

Time it took to move the file: 744.54 seconds

CYBERSECURITY REPORT – LABORATORY 9

- Observe and analyze the communication process and its content in the WS.
- Measure the time of data transfer.

Configuration	AES-256-GCM - SHA512	AES-128-CBC - SHA1	DES-EDE-CBC - MD5	BF -CBC - MD5
Time (seconds)	872.92	610.9	611.23	744.54

4.4 Is there a difference in data transfer times? What can affect them?

Yes, times differ:

- AES-256-GCM / SHA512 → 872.92 s
- AES-128-CBC / SHA1 → 610.90 s
- DES-EDE-CBC / MD5 → 611.23 s
- BF-CBC / MD5 → 744.54 s

Factors affecting times:

1. Cipher computational cost (AES-256 > AES-128 > DES/Blowfish)
2. Authentication overhead (SHA512 > SHA1 > MD5)
3. Mode differences (GCM vs CBC)
4. Packet size, fragmentation, retransmissions
5. CPU hardware acceleration, disk I/O, system load

4.5 What can we say about the encryption and authentication algorithms used?

AES-256-GCM / SHA512 → Very strong, modern AEAD

AES-128-CBC / SHA1 → Secure cipher, but SHA1 is weak for integrity

DES-EDE-CBC / MD5 → Weak, slow, outdated

BF-CBC / MD5 → Weak, small block size, insecure hash

CYBERSECURITY REPORT – LABORATORY 9

4.6 Which set of settings is the worst and which is the best in terms of security?

AES-256-GCM + SHA512 

AES-128-CBC + SHA1 

DES-EDE-CBC + MD5 

BF-CBC + MD5 

Avoid MD5/SHA1 and 64-bit block ciphers (DES/Blowfish)

AES-GCM (AEAD) is preferred; AES-128 is OK if performance is critical